Practical-week-7

Task-1:

Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
arikela@DESKTOP-AMUSJME:

  GNU nano 8.7.1                        task1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;

void code_function()
{
    printf("Inside code_function()\n");
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Heap allocation failed\n");
        return 1;
    }

    *heap_var = 40;

    printf("\n===== MEMORY ADDRESSES =====\n");

    printf("Code address   : %p\n", (void *)code_function);
    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address   : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);
                           [ Read 43 lines ]
^G Help       ^O Write Out^F Where Is ^K Cut        ^T Execute
^X Exit       ^R Read File^\ Replace  ^U Paste      ^J Justify
```

arikela@DESKTOP-AMUSJME:

```
  GNU nano 8.7.1                task1.c
int main()
{
    static int static_var = 20;
    int stack_var = 30;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Heap allocation failed\n");
        return 1;
    }

    *heap_var = 40;

    printf("\n===== MEMORY ADDRESSES =====\n");

    printf("Code address   : %p\n", (void *)code_function);
    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address   : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);

    printf("\nProcess ID: %d\n", getpid());

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}

^G Help      ^O Write Out^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File^\ Replace   ^U Paste     ^J Justify
```

```
arikela@DESKTOP-AMUSJME:~/task1_memory$ gcc task1.c -o task1
arikela@DESKTOP-AMUSJME:~/task1_memory$ ls
task1  task1.c
arikela@DESKTOP-AMUSJME:~/task1_memory$ ./task1

===== MEMORY ADDRESSES =====
Code address    : 0x637722adf209
Global address : 0x637722ae2010
Static address : 0x637722ae2014
Heap address    : 0x63772599d010
Stack address   : 0x7ffec1022b2c

Process ID: 764

Press Enter to exit...
```

In second terminal:

```
arikela@DESKTOP-AMUSJME:
arikela@DESKTOP-AMUSJME:~$ ps -ef | grep task1
arikela      764     348  0 03:47 pts/0    00:00:00 ./task1
arikela      818     781  0 03:50 pts/2    00:00:00 grep --color=auto task1
arikela@DESKTOP-AMUSJME:~$ pidof task1
764
arikela@DESKTOP-AMUSJME:~$ cat /proc/764/maps
637722ade000-637722adf000 r--p 00000000 08:30 39189                      /home/arikela/task1_memory/task1
637722adf000-637722ae0000 r-xp 00001000 08:30 39189                      /home/arikela/task1_memory/task1
637722ae0000-637722ae1000 r--p 00002000 08:30 39189                      /home/arikela/task1_memory/task1
637722ae1000-637722ae2000 r--p 00002000 08:30 39189                      /home/arikela/task1_memory/task1
637722ae2000-637722ae3000 rw-p 00003000 08:30 39189                      /home/arikela/task1_memory/task1
63772599d000-6377259be000 rw-p 00000000 00:00 0                          [heap]
7f9c5a800000-7f9c5a828000 r--p 00000000 08:30 13628                      /usr/lib/x86_64-linux-gnu/libc.so.6
7f9c5a828000-7f9c5a9c0000 r-xp 00028000 08:30 13628                      /usr/lib/x86_64-linux-gnu/libc.so.6
7f9c5a9c0000-7f9c5aa0e000 r--p 001c0000 08:30 13628                      /usr/lib/x86_64-linux-gnu/libc.so.6
7f9c5aa0e000-7f9c5aa12000 r--p 0020d000 08:30 13628                      /usr/lib/x86_64-linux-gnu/libc.so.6
7f9c5aa12000-7f9c5aa14000 rw-p 00211000 08:30 13628                      /usr/lib/x86_64-linux-gnu/libc.so.6
7f9c5aa14000-7f9c5aa21000 rw-p 00000000 00:00 0
7f9c5aa5e000-7f9c5aa61000 rw-p 00000000 00:00 0
7f9c5aa67000-7f9c5aa69000 rw-p 00000000 00:00 0
7f9c5aa69000-7f9c5aa6d000 r--p 00000000 00:00 0                          [vvar]
7f9c5aa6d000-7f9c5aa6f000 r--p 00000000 00:00 0                          [vvar_vclock]
7f9c5aa6f000-7f9c5aa71000 r-xp 00000000 00:00 0                          [vdso]
7f9c5aa71000-7f9c5aa72000 r--p 00000000 08:30 13625                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f9c5aa72000-7f9c5aaa1000 r-xp 00001000 08:30 13625                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f9c5aaa1000-7f9c5aaac000 r--p 00030000 08:30 13625                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f9c5aaac000-7f9c5aaae000 r--p 0003b000 08:30 13625                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f9c5aaae000-7f9c5aaaf000 rw-p 0003d000 08:30 13625                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7f9c5aaaf000-7f9c5aab0000 rw-p 00000000 00:00 0
7ffec1004000-7ffec1025000 rw-p 00000000 00:00 0                          [stack]
arikela@DESKTOP-AMUSJME:~$
```

```
arikela@DESKTOP-AMUSJME:~$ cat /proc/764/maps | grep heap
63772599d000-6377259be000 rw-p 00000000 00:00 0                          [heap]
```

Back to terminal 1:

```
arikela@DESKTOP-AMUSJME:~/task1_memory$ ./task1

===== MEMORY ADDRESSES =====
Code address   : 0x637722adf209
Global address : 0x637722ae2010
Static address : 0x637722ae2014
Heap address   : 0x63772599d010
Stack address  : 0x7ffec1022b2c

Process ID: 764

Press Enter to exit...

arikela@DESKTOP-AMUSJME:~/task1_memory$ ./task1

===== MEMORY ADDRESSES =====
Code address   : 0x576f95d81209
Global address : 0x576f95d84010
Static address : 0x576f95d84014
Heap address   : 0x576fb87c1010
Stack address  : 0x7ffe357cfdfc

Process ID: 910

Press Enter to exit...
```

Task-2:

Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

In terminal 1:

```
arikela@DESKTOP-AMUSJME:~/task1_memory$ ./task1

===== MEMORY ADDRESSES =====
Code address   : 0x576f95d81209
Global address : 0x576f95d84010
Static address : 0x576f95d84014
Heap address   : 0x576fb87c1010
Stack address  : 0x7ffe357cfdfc

Process ID: 910

Press Enter to exit...
```

In terminal 2:

```
arikela@DESKTOP-AMUSJME
arikela@DESKTOP-AMUSJME:~/task1_memory$ pidof task1
910
arikela@DESKTOP-AMUSJME:~/task1_memory$ cat /proc/910/maps
576f95d80000-576f95d81000 r--p 00000000 08:30 39189                      /home/arikela/task1_m
ory/task1
576f95d81000-576f95d82000 r-xp 00001000 08:30 39189                      /home/arikela/task1_m
ory/task1
576f95d82000-576f95d83000 r--p 00002000 08:30 39189                      /home/arikela/task1_m
ory/task1
576f95d83000-576f95d84000 r--p 00002000 08:30 39189                      /home/arikela/task1_m
ory/task1
576f95d84000-576f95d85000 rw-p 00003000 08:30 39189                      /home/arikela/task1_m
ory/task1
576fb87c1000-576fb87e2000 rw-p 00000000 00:00 0                          [heap]
70acbe400000-70acbe428000 r--p 00000000 08:30 13628                      /usr/lib/x86_64-linux
nu/libc.so.6
70acbe428000-70acbe5c0000 r-xp 00028000 08:30 13628                      /usr/lib/x86_64-linux
nu/libc.so.6
70acbe5c0000-70acbe60e000 r--p 001c0000 08:30 13628                      /usr/lib/x86_64-linux
nu/libc.so.6
70acbe60e000-70acbe612000 r--p 0020d000 08:30 13628                      /usr/lib/x86_64-linux
nu/libc.so.6
70acbe612000-70acbe614000 rw-p 00211000 08:30 13628                      /usr/lib/x86_64-linux
nu/libc.so.6
70acbe614000-70acbe621000 rw-p 00000000 00:00 0
70acbe771000-70acbe774000 rw-p 00000000 00:00 0
70acbe77a000-70acbe77c000 rw-p 00000000 00:00 0
70acbe77c000-70acbe780000 r--p 00000000 00:00 0                          [vvar]
70acbe780000-70acbe782000 r--p 00000000 00:00 0                          [vvar_vclock]
70acbe782000-70acbe784000 r-xp 00000000 00:00 0                          [vdso]
```

```
arikela@DESKTOP-AMUSJME:~/task1_memory$ cat /proc/$(pidof task1)/maps | grep task1
576f95d80000-576f95d81000 r--p 00000000 08:30 39189                      /home/arikela/task1_memory/task1
576f95d81000-576f95d82000 r-xp 00001000 08:30 39189                      /home/arikela/task1_memory/task1
576f95d82000-576f95d83000 r--p 00002000 08:30 39189                      /home/arikela/task1_memory/task1
576f95d83000-576f95d84000 r--p 00002000 08:30 39189                      /home/arikela/task1_memory/task1
576f95d84000-576f95d85000 rw-p 00003000 08:30 39189                      /home/arikela/task1_memory/task1
arikela@DESKTOP-AMUSJME:~/task1_memory$ pmap $(pidof task1)
910:   ./task1
0000576f95d80000      4K r---- task1
0000576f95d81000      4K r-x-- task1
0000576f95d82000      4K r---- task1
0000576f95d83000      4K r---- task1
0000576f95d84000      4K rw--- task1
0000576fb87c1000    132K rw---   [ anon ]
000070acbe400000    160K r---- libc.so.6
000070acbe428000   1632K r-x-- libc.so.6
000070acbe5c0000    312K r---- libc.so.6
000070acbe60e000     16K r---- libc.so.6
000070acbe612000      8K rw--- libc.so.6
000070acbe614000     52K rw---   [ anon ]
000070acbe771000     12K rw---   [ anon ]
000070acbe77a000      8K rw---   [ anon ]
000070acbe77c000     16K r----   [ anon ]
000070acbe780000      8K r----   [ anon ]
000070acbe782000      8K r-x--   [ anon ]
```

```
arikela@DESKTOP-AMUSJME:~/task1_memory$ pmap -x $(pidof task1)
910:   ./task1
Address           Kbytes     RSS   Dirty Mode  Mapping
0000576f95d80000       4       4       0 r---- task1
0000576f95d81000       4       4       0 r-x-- task1
0000576f95d82000       4       4       0 r---- task1
0000576f95d83000       4       4       4 r---- task1
0000576f95d84000       4       4       4 rw--- task1
0000576fb87c1000     132       4       4 rw---   [ anon ]
000070acbe400000     160     160       0 r---- libc.so.6
000070acbe428000    1632    1120       0 r-x-- libc.so.6
000070acbe5c0000     312     184       0 r---- libc.so.6
000070acbe60e000      16      16      16 r---- libc.so.6
000070acbe612000       8       8       8 rw--- libc.so.6
000070acbe614000      52      20      20 rw---   [ anon ]
000070acbe771000      12       8       8 rw---   [ anon ]
000070acbe77a000       8       4       4 rw---   [ anon ]
000070acbe77c000      16       0       0 r----   [ anon ]
000070acbe780000       8       0       0 r----   [ anon ]
000070acbe782000       8       8       0 r-x--   [ anon ]
000070acbe784000       4       4       0 r---- ld-linux-x86-64.so.2
000070acbe785000     188     188       0 r-x-- ld-linux-x86-64.so.2
000070acbe7b4000      44      44       0 r---- ld-linux-x86-64.so.2
000070acbe7bf000       8       8       8 r---- ld-linux-x86-64.so.2
000070acbe7c1000       4       4       4 rw--- ld-linux-x86-64.so.2
000070acbe7c2000       4       4       4 rw---   [ anon ]
00007ffe357b0000     132      12      12 rw---   [ stack ]
---------------- ------- ------- -------
total kB            2768    1816      96
```